Spectral asymptotics and converses for accelerated quadratic iterations

Abstract

Bach conjectures three pointwise asymptotic regimes for a discrete accelerated quadratic iteration with exact power-law spectral data. We prove the stated equivalents and constants for every fixed positive real momentum parameter and every positive spectral dimension in the finite-energy formulation. A fixed asymptotically equivalent monotone spectrum shows that the Stieltjes data alone do not determine the pointwise constant. We then give sharp perturbation-stability thresholds and an exact interval for the possible logarithmic-boundary coefficients, attained by a fixed monotone spectrum. For zero-start Nesterov regularization on an arbitrary Hilbert-space operator, we characterize all-iteration norm rates strictly above semisaturation by weak spectral tails, giving the stronger source index asked for by Kindermann with the classical range-endpoint epsilon loss. At semisaturation itself, the exact condition is bounded weighted mass on moving logarithmic spectral intervals. This condition lies strictly between the strong source condition and the weak spectral-tail bound. The converse proofs use uniform averages of the actual squared residuals, including the compact Bessel transition.

Keywords: acceleration; spectral asymptotics; stability; semisaturation

2020 Mathematics Subject Classification. Primary 65K10; Secondary 65J20, 47B15.

1 Introduction

Fix

$$\rho, \alpha, L, \Delta, \gamma > 0, \qquad \tau = \gamma L \in (0,1), \qquad \omega > 0, \qquad \beta = 1 + \alpha(\omega - 1).$$

We first use the exact normalized eigenvalues

$$\lambda_j = \tau j^{-\alpha}, \qquad x_{0,j} = x_{1,j} = \Delta j^{-\beta/2}, \qquad j \geq 1, \tag{1}$$

and the coordinate recurrence

$$x_{k+1,j} = (1 - \lambda_j)\left[x_{k,j} + \frac{k-1}{k+2\rho-1}(x_{k,j} - x_{k-1,j})\right], \qquad k \geq 1. \tag{2}$$

Its criterion and normalization are

$$a_k = \frac{1}{2\gamma}\sum_{j\geq 1} \lambda_j x_{k,j}^2, \qquad c = \frac{L\Delta^2}{2\alpha\tau^\omega}. \tag{3}$$

The condition $\omega > 0$, or equivalently $\alpha + \beta > 1$, makes the initial energy finite. If $\beta > 1$, the initial state lies in the usual ℓ^2 space and (2) is the Hilbert-space iteration with

diagonal operator $He_j = Lj^{-\alpha}e_j$. For $\beta \le 1$ we retain the coordinatewise finite-energy meaning. This is the convention explicitly allowed in Bach [4, pp. 8–10]; it does not assert ordinary ℓ^2 membership.

Bach's Conjecture 11 gives three exact asymptotic formulas for (2) when ρ is a positive integer, and proposes the extension to all $\rho > 0$. His Proposition 9 proves the case $\rho = 1$. For larger integer parameters, generating-function calculations give the proposed constants, but the pointwise justification is left open. We establish the equivalents directly on the exact model (1).

The polynomial representation and spectral transition mechanism have important predecessors. Kindermann [10] gives a Gegenbauer residual for a closely related inverse-problem iteration, with a different initialization. Jacobi-polynomial iterations occur in accelerated gossip [5]. Cunha et al. [7] prove three-regime asymptotics for ordinary Nesterov damping under continuous expected spectral-density assumptions. Bach's direct proof for $\rho = 1$ already combines a Bessel-scale Riemann sum, a damping-scale oscillatory sum, and a widening harmonic window at the boundary [4, Appendix B]. We extend that exact-grid argument using the classical uniform Hilb approximation for Jacobi polynomials [8]. The new pointwise conclusions rely on uniform remainder bounds and cancellation on the actual integer spectrum, not on replacing it by a density.

Define the positive constants

$$G_\rho = \frac{\Gamma(2\rho)^2}{2^{2\rho-1}\Gamma(\rho)^2}, \tag{4}$$

and, for $0 < \omega < \rho$,

$$M_{\rho,\omega} = \frac{2^{2\omega}\Gamma(\rho+1/2)^2\Gamma(2\rho-2\omega)\Gamma(\omega)}{\Gamma(\rho-\omega+1/2)^2\Gamma(2\rho-\omega)}. \tag{5}$$

All gamma arguments in (4)–(5) are positive on their stated domains.

Theorem 1. *For every fixed $\rho, \alpha, \omega > 0$ and the data (1), recurrence (2) has finite criterion (3) at every iteration, and*

$$a_k \sim c \begin{cases} M_{\rho,\omega}k^{-2\omega}, & 0 < \omega < \rho, \\ G_\rho(\log k)k^{-2\rho}, & \omega = \rho, \\ G_\rho\Gamma(\omega-\rho)k^{-\omega-\rho}, & \omega > \rho. \end{cases} \tag{6}$$

The constants agree with Bach's displayed formulas wherever the latter are defined. The nonsingular form (5) also gives the continuous interpretation at the removable simultaneous poles in the proposed real-ρ formula; we verify this at the end of the proof.

The exact spectral data matter. To formulate the limitation precisely, for a finite positive measure σ supported in $[0, 1)$ put

$$S(u) = \int \frac{d\sigma(\lambda)}{\lambda + u}, \qquad u > 0.$$

Bach's assumption (A1) is that, for every integer $\ell > \omega$,

$$S^{(\ell-1)}(u) \sim c(-1)^{\ell-1}\Gamma(\omega)\Gamma(\ell-\omega)u^{\omega-\ell} \quad \text{as } u \downarrow 0. \tag{7}$$

His Lemma 7 obtains this condition from exact spectral and coordinate powers. Conjecture 11 uses the qualifier "Assume (A1) obtained from Lemma 7." The next theorem distinguishes that exact-grid interpretation from an enlargement using (7) alone.

Theorem 2. *Fix $\alpha, L, \Delta, \gamma > 0$, $\tau = \gamma L \in (0,1)$, and $\omega > 2$. Set $\beta = 1 + \alpha(\omega-1)$ and c as in* (3)*. There exists one fixed positive nonincreasing sequence*

$$\widetilde{\lambda}_1 = \tau, \qquad \widetilde{\lambda}_j \sim \tau j^{-\alpha},$$

such that recurrence (2) *with $\rho = 2$, eigenvalues $\widetilde{\lambda}_j$, and the same initial coordinates satisfies*

$$\liminf_{k\to\infty} k^{\omega+2}a_k = 0, \qquad \limsup_{k\to\infty} k^{\omega+2}a_k = 9c\Gamma(\omega-2). \tag{8}$$

Here the criterion uses the actual eigenvalues $\widetilde{\lambda}_j$. Its performance measure

$$\sigma = \frac{\Delta^2}{2\gamma}\sum_{j\ge 1} j^{-\beta}\widetilde{\lambda}_j\,\delta_{\widetilde{\lambda}_j} \tag{9}$$

satisfies (7) *with the same ω, c. The corresponding diagonal operator is compact and bounded, and the initial state belongs to ordinary ℓ^2.*

The upper value in (8) is twice the exact-grid coefficient in (6). This is not a counterexample to the exact-grid statement. It shows that asymptotically equivalent monotone eigenvalues and the transform data (7) do not determine the pointwise constant. Spectral placement and gluing constructions already occur in Velikanov and Yarotsky [13, Theorem 4.11 and Appendix G.1]; we use a phase-specific construction to obtain the two exact extrema for the present recurrence.

The loss of a pointwise constant can be quantified by the size of the eigenvalue error. Section 6 gives sharp stability thresholds and fixed monotone examples for relative power-size perturbations. In particular, the logarithmic boundary admits a sharp coefficient interval below its inclusive stability threshold. These conclusions retain exact initial coordinate powers and actual objective weights.

The loss of an exact pointwise constant does not prevent a characterization of all-iteration norm rates. We now use a different initialization, namely the zero-start inverse-problem iteration considered by Kindermann [10]. Let $T : X \to Y$ be bounded between real Hilbert spaces, put $Q = T^*T$, assume $\|Q\| \le 1$, and let $u^\dagger \perp \ker T$ with exact data $y = Tu^\dagger$. For fixed $\rho > 0$, set

$$\begin{aligned} u_0 &= 0, \qquad u_1 = T^*y, \\ v_k &= u_k + \frac{k-1}{k+2\rho-1}(u_k - u_{k-1}), \\ u_{k+1} &= v_k + T^*(y - Tv_k), \qquad k \ge 1. \end{aligned} \tag{10}$$

Write E_Q for the spectral projection measure of Q and $\mu(B) = \|E_Q(B)u^\dagger\|^2$.

Theorem 3. *For every fixed $\rho > 0$ and $\xi > \rho$, iteration* (10) *satisfies*

$$\|u_k - u^\dagger\| = O(k^{-\xi}) \quad \Longleftrightarrow \quad \mu((0,t]) = O(t^{2\xi-\rho}) \quad (t \downarrow 0). \tag{11}$$

The rate on the left holds at every sufficiently large integer iteration. Consequently,

$$u^\dagger \in \operatorname{Ran} Q^{\xi-\rho/2-\epsilon} \qquad \text{for every } 0 < \epsilon < \xi - \rho/2. \tag{12}$$

The endpoint membership $u^\dagger \in \operatorname{Ran} Q^{\xi-\rho/2}$ does not follow in general.

No compactness, eigenvalue-grid, density, injectivity, or positive-gap assumption is imposed in this theorem. Kindermann's momentum parameter is $2\rho - 1$ in our notation. His Theorem 5 gives the index $\xi/2 - \epsilon$, and Remark 2 asks for the stronger index $\xi - \rho/2$ when $\xi > \rho$. Theorem 3 gives the corresponding weak-tail characterization and the usual epsilon-loss range consequence under precisely the noisefree, zero-start, all-iteration hypothesis.

Direct and converse results for polynomial semiiterations have a substantial history, including Hanke [9]. General regularization frameworks may also admit oscillatory residuals [6]. Shrinking-angle Jacobi estimates also appear in recent Nesterov analysis [11, Proposition 2.3]. The weak spectral spaces, their interpolation interpretation, and the loss at a classical range endpoint are established [1, 2]. Our proof checks the needed temporal lower bound for the actual Landweber-damped Gegenbauer filter. It does not infer a lower bound from an upper envelope or from a pointwise approximation at possible zeros. No noisy stopping-rule equivalence is asserted. The critical case $\xi = \rho$ has a different characterization: Section 8 proves an exact weighted logarithmic-annulus condition, strictly between the strong source and weak-tail conditions.

We prove uniform scalar estimates in Section 2 and the lattice estimate in Section 3. Sections 4 and 5 prove Theorems 1 and 2. Section 6 establishes the perturbation thresholds and sharp examples. Section 7 establishes the temporal-window estimate and proves Theorem 3. Section 8 treats the exact semisaturation endpoint.

2 Scalar residuals and uniform polynomial estimates

Let $p_0(\lambda) = p_1(\lambda) = 1$ and

$$(k + 2\rho - 1)p_{k+1}(\lambda) = (1 - \lambda)\{2(k + \rho - 1)p_k(\lambda) - (k - 1)p_{k-1}(\lambda)\}. \tag{13}$$

For fixed k, this is a polynomial in λ. To justify the finite-energy recurrence, define

$$\mathcal{E} = \left\{ x : \sum_{j\geq 1} \lambda_j x_j^2 < \infty \right\}, \qquad \|x\|_{\mathcal{E}}^2 = \sum_{j\geq 1} \lambda_j x_j^2.$$

Multiplication by $\sqrt{\lambda_j}$ is an isometry onto ℓ^2. The diagonal operator H with entries λ_j/γ is bounded on $\mathcal{E}$, and

$$\|x_0\|_{\mathcal{E}}^2 = \tau\Delta^2 \sum_{j\geq 1} j^{-1-\alpha\omega} < \infty.$$

Thus (2) defines every iterate in $\mathcal{E}$ and $x_{k,j} = p_k(\lambda_j)x_{0,j}$. This is a bounded linear recurrence on $\mathcal{E}$; we do not identify it with a gradient iteration for the $\mathcal{E}$ inner product. With $A = L\Delta^2/2$ and $q_j = 1-\lambda_j$, its exact energy is

$$a_k = A\sum_{j\geq 1} j^{-1-\alpha\omega} p_k(\lambda_j)^2. \tag{14}$$

This series converges for every fixed k because p_k is bounded on $[0,\tau]$.

Put

$$q = 1-\lambda, \quad \theta = \arcsin\sqrt{\lambda}, \quad n = k-1, \quad N = k+\rho-1, \quad \nu = \rho - \tfrac{1}{2}.$$

We use the normalized Bessel profile

$$\Phi_\rho(t) = \Gamma(\rho+1/2)(2/t)^{\rho-1/2} J_{\rho-1/2}(t), \qquad \Phi_\rho(0) = 1, \tag{15}$$

and

$$D_\rho = \frac{2^\rho\Gamma(\rho+1/2)}{\sqrt{\pi}} = \frac{2^{1-\rho}\Gamma(2\rho)}{\Gamma(\rho)}, \qquad D_\rho^2/2 = G_\rho. \tag{16}$$

Lemma 1. *For $k \geq 1$ and $0 < \lambda \leq \tau$,*

$$p_k(\lambda) = q^{(k-1)/2}\frac{C_{k-1}^\rho(\cos\theta)}{C_{k-1}^\rho(1)} = q^{(k-1)/2}\frac{P_{k-1}^{(\nu,\nu)}(\cos\theta)}{P_{k-1}^{(\nu,\nu)}(1)}. \tag{17}$$

There is a constant C, depending only on fixed ρ,τ, such that

$$|p_k(\lambda)| \leq Cq^{(k-1)/2}\min\{1,(k\sqrt{\lambda})^{-\rho}\}. \tag{18}$$

Uniformly for t in compact subsets of $(0,\infty)$,

$$p_k(t^2/k^2) \longrightarrow \Phi_\rho(t). \tag{19}$$

For each fixed $t_0 > 0$, uniformly when $k\sqrt{\lambda} \geq t_0$,

$$p_k(\lambda) = D_\rho N^{-\rho}\lambda^{-\rho/2}q^{(k-1)/2}\left[\cos(N\theta - \rho\pi/2) + \varepsilon_k(\lambda)\right], \quad |\varepsilon_k(\lambda)| \leq \frac{C_{t_0}}{k\sqrt{\lambda}}. \tag{20}$$

Proof. The normalized Gegenbauer recurrence is

$$(n+2\rho)R_{n+1}(x) = 2(n+\rho)xR_n(x) - nR_{n-1}(x), \qquad R_n(x) = \frac{C_n^\rho(x)}{C_n^\rho(1)}.$$

Multiplication by the appropriate powers of $q^{1/2}$ proves (13). The first values are $p_1 = 1$, $p_2 = q$. The p_0 coefficient in the first step is zero, so the initialization is exact. The

Jacobi equality follows from the standard proportionality of these polynomials and their normalization at one.

We use the classical Hilb formula as stated in Frenzen and Wong [8, Eq. (1.1), p. 979]. For fixed $a > -1$, real b, and $N_n = n + (a + b + 1)/2$,

$$\left(\sin\frac{\theta}{2}\right)^a \left(\cos\frac{\theta}{2}\right)^b P_n^{(a,b)}(\cos\theta) = N_n^{-a}\frac{\Gamma(n+a+1)}{n!}\left(\frac{\theta}{\sin\theta}\right)^{1/2} J_a(N_n\theta) + \mathcal{R}_n(\theta), \tag{21}$$

where, for fixed positive c_0, ϵ,

$$\mathcal{R}_n(\theta) = \begin{cases} \theta^{1/2}O(n^{-3/2}), & c_0/n \le \theta \le \pi - \epsilon, \\ \theta^{a+2}O(n^a), & 0 \le \theta \le c_0/n. \end{cases} \tag{22}$$

Both bounds are uniform on the indicated ranges. Here $a = b = \nu = \rho - 1/2 > -1/2$, so the hypothesis holds for every $\rho > 0$. We only need $\theta \le \arcsin\sqrt{\tau} < \pi/2$.

Divide (21) by $P_n^{(\nu,\nu)}(1) = \Gamma(n+\nu+1)/(\Gamma(\nu+1)n!)$ and by its trigonometric factor. The leading gamma ratio cancels exactly. We obtain

$$R_n(\cos\theta) = \left(\frac{\theta}{\sin\theta}\right)^\rho \Phi_\rho(N\theta) + E_n(\theta), \tag{23}$$

with

$$|E_n(\theta)| \le \begin{cases} Cn^{-\rho-1}\theta^{1-\rho}, & \theta \ge c_0/n, \\ C\theta^2, & \theta \le c_0/n. \end{cases} \tag{24}$$

The fixed gamma-ratio bounds in the error give these estimates; n and N are comparable for large k.

For $\nu > -1/2$, the classical Bessel bounds give $J_\nu(t) = O(t^\nu)$ at zero and

$$J_\nu(t) = \sqrt{\frac{2}{\pi t}}\left[\cos(t - \nu\pi/2 - \pi/4) + O(t^{-1})\right] \tag{25}$$

at infinity [12, Sections 10.7 and 10.17]. Even when $\nu < 0$, the factor $t^{-\nu}$ in (15) cancels the singularity at zero. Hence $|\Phi_\rho(t)| \le C\min\{1, t^{-\rho}\}$. On the large-angle range the error in (24), divided by this envelope, is $O(\theta/n)$. On the small-angle range it is $O(\theta^2)$. This proves (18). It also proves the compact limit (19), retaining the factor $q^{(k-1)/2} \to 1$.

For (20), enlarge the remainder constant in (25) to all $N\theta$ above a fixed positive threshold. Its leading amplitude in (23) is exactly $D_\rho N^{-\rho}(\sin\theta)^{-\rho}$, and its phase is $N\theta - \rho\pi/2$. The relative envelope error from (24) is $O(\theta/n)$, which is bounded by $C/(k\sqrt{\lambda})$ on the fixed angle interval. The Bessel error has the same bound. This is an absolute error inside the cosine bracket, and remains valid at its zeros. Finitely many small k are absorbed into the constants. ∎

For later use at $\rho = 2$, an exact version is available. Differentiating $U_k(\cos\theta) = \sin((k+1)\theta)/\sin\theta$ and using $U_k' = 2C_{k-1}^2$ gives

$$p_k(\lambda) = -\frac{3q^{(k-1)/2}}{k(k+2)\lambda} B_k(\lambda), \quad B_k(\lambda) = \cos((k+1)\theta) - \frac{\cot\theta}{k+1}\sin((k+1)\theta). \tag{26}$$

Since $B_k = \cos t - (\theta\cot\theta)\sin t/t$, with $t = (k+1)\theta$ and $0 < \theta\cot\theta \le 1$, one has

$$|B_k(\lambda)| \le 2. \tag{27}$$

3 Uniform cancellation on the integer spectrum

Let $\theta_j = \arcsin\sqrt{\tau j^{-\alpha}}$. The frequency below is $N_k = k + \rho - 1$, with $\rho > 0$ fixed. Constants may depend on fixed α, τ, ρ.

Lemma 2. *Choose an integer $d \ge 3$ with $d > \alpha/2$, and put $D = 2^d$. For $M \ge 1$ and large k,*

$$\begin{aligned}\sup_{M\le u\le v\le 2M}\left|\sum_{u<j\le v} e^{2\mathrm{i}N_k\theta_j}\right| \\ \le CM\left\{M^{-2/D} + (kM^{-\alpha/2-d})^{1/(D-2)} + (kM^{-\alpha/2})^{-2/D}\right\}.\end{aligned} \tag{28}$$

It is $o(M)$ uniformly when $k^{1/\alpha} \le M \le k^{(2-\eta)/\alpha}$ for fixed $\eta \in (0,1)$. On every fixed scaled interval $[\varepsilon k^{1/\alpha}, Rk^{1/\alpha}]$, all partial sums are uniformly $o(k^{1/\alpha})$.

Proof. Set $a = \alpha/2$ and

$$f_k(x) = \frac{N_k}{\pi}\arcsin(\sqrt{\tau}x^{-a}).$$

The positive power series $\arcsin z = \sum_{r\ge0} b_r z^{2r+1}$, where $b_r = \binom{2r}{r}/(4^r(2r+1))$, may be differentiated a fixed number of times uniformly for $x \ge 1$, because $\tau < 1$. Thus

$$(-1)^d f_k^{(d)}(x) = \frac{N_k\sqrt{\tau}}{\pi} x^{-a-d}\sum_{r\ge0} b_r\tau^r((2r+1)a)_d x^{-2ra}.$$

The first term is positive and the whole series is bounded. On $[M, 2M]$ this gives two-sided positive bounds comparable to $h_M = kM^{-a-d}$.

Theorem 11 of Arias de Reyna [3] applies on an interval of length Y when $\lfloor Y\rfloor > d$ and $0 < \ell \le g^{(d)} \le \Lambda$. Its normalized exponential-sum bound is

$$C_d \max\left\{\left(\frac{\Lambda}{\ell Y}\right)^{2/D}, \left(\frac{\Lambda^2}{\ell}\right)^{1/(D-2)}, (\ell Y^d)^{-2/D}\right\}.$$

Use $g = (-1)^d f_k$. For a partial block, multiplication by Y gives

$$C\max\{Y^{1-2/D},\ Yh_M^{1/(D-2)},\ h_M^{-2/D}Y^{1-2d/D}\}.$$

Every power of Y is nonnegative, since $2d \le 2^d$. Replacing $Y \le M$ by M proves (28). For $\lfloor Y \rfloor \le d$, the trivial bound $d+1$ is absorbed by its first term. This establishes the estimate for all partial intervals before summation by parts.

On the widening range the three terms in braces are bounded by

$$k^{-2/(\alpha D)}, \quad Ck^{-(d/\alpha-1/2)/(D-2)}, \quad k^{-\eta/D},$$

respectively. Their exponents are negative. On a fixed scaled interval all block lengths are comparable to $k^{1/\alpha}$, so the same conclusion follows by a finite dyadic cover. ∎

A weight whose supremum and total variation are $O(M^{-1})$ converts an $o(M)$ partial-sum bound into a weighted $o(1)$ estimate by summation by parts. Constant phase shifts do not affect Lemma 2.

4 Proof of Theorem 1

4.1 The regime $0 < \omega < \rho$

Set $r = k^{2/\alpha}$. Equation (14) gives

$$k^{2\omega} a_k = \frac{A}{r} \sum_{j \ge 1} (j/r)^{-1-\alpha\omega} p_k(\tau k^{-2} (j/r)^{-\alpha})^2.$$

On compact positive x intervals, Lemma 1 gives the uniform limit $\Phi_\rho(\sqrt{\tau} x^{-\alpha/2})^2$ for the scalar factor. The global envelope is

$$C x^{-1-\alpha\omega} \min\{1, x^{\alpha\rho}\}.$$

At zero its exponent is $-1 + \alpha(\rho - \omega) > -1$; at infinity it is $-1 - \alpha\omega < -1$. Ordinary power-sum comparison bounds the mesh tails by $C\varepsilon^{\alpha(\rho-\omega)}$ and $CR^{-\alpha\omega}$, uniformly in k. Compact Riemann-sum convergence followed by removal of both cutoffs gives

$$\begin{aligned} \lim_{k\to\infty} k^{2\omega} a_k &= A \int_0^\infty x^{-1-\alpha\omega} \Phi_\rho(\sqrt{\tau} x^{-\alpha/2})^2 \, dx \\ &= 2c \int_0^\infty t^{2\omega-1} \Phi_\rho(t)^2 \, dt. \end{aligned} \tag{29}$$

This includes $0 < \omega \le 1$ without any ordinary ℓ^2 assumption.

The equal-argument Weber–Schafheitlin formula [12, Eq. 10.22.57] is

$$\int_0^\infty t^{-\ell} J_\nu(t)^2 \, dt = \frac{\Gamma(\ell)\Gamma(\nu + (1-\ell)/2)}{2^\ell \Gamma((\ell+1)/2)^2 \Gamma(\nu + (1+\ell)/2)}, \qquad 0 < \ell < 2\nu + 1.$$

Take $\nu = \rho - 1/2$ and $\ell = 2\rho - 2\omega$. Its strict inequalities are exactly $0 < \omega < \rho$. Substitution into (29) gives $cM_{\rho,\omega}$. Negative Bessel orders when $\rho < 1/2$ cause no exception, because the normalized endpoint behavior was already included in Lemma 1.

4.2 The boundary $\omega = \rho$

Put $n = k^{1/\alpha}$ and $N_k = k + \rho - 1$. The envelope (18) gives

$$A \sum_{j \le n} j^{-1-\alpha\rho} p_k(\lambda_j)^2 \le C k^{-2\rho} \sum_{j \le n} j^{-1} \exp[-(\tau/2)(n/j)^\alpha] = O(k^{-2\rho}).$$

The harmonic sum is bounded by splitting into dyadic blocks below n; their dampings form the summable sequence $\exp[-(\tau/2)2^{h\alpha}]$. The upper tail satisfies

$$A \sum_{j > n^2} j^{-1-\alpha\rho} p_k(\lambda_j)^2 \le C \sum_{j > n^2} j^{-1-\alpha\rho} = O(k^{-2\rho}),$$

since $\alpha\rho > 0$.

On $n < j \le n^2$, formula (20) applies with a fixed positive threshold below $\sqrt{\tau}$. Its bracket error is bounded by $C j^{\alpha/2}/k$. The cross and squared errors summed with weight $j^{-1} q_j^{k-1}$ are $O(1)$, because

$$k^{-1} \sum_{j \le n^2} j^{\alpha/2-1} = O(1), \qquad k^{-2} \sum_{j \le n^2} j^{\alpha-1} = O(1).$$

Consequently,

$$a_k = A D_\rho^2 N_k^{-2\rho} \tau^{-\rho} \times \left[\sum_{n < j \le n^2} \frac{q_j^{k-1}}{j} \cos^2(N_k \theta_j - \rho\pi/2) + O(1) \right]. \tag{30}$$

The previously discarded tails can be included in the bounded remainder. Bernoulli's inequality yields

$$\sum_{j > n} \frac{1 - q_j^{k-1}}{j} \le \tau k \sum_{j > n} j^{-1-\alpha} = O(1).$$

Thus damping can be replaced by one at bounded cost, and the nonoscillatory harmonic mass is

$$\sum_{n < j \le n^2} \frac{1}{j} = \log n + O(1).$$

For fixed $\eta \in (0, 1)$, Lemma 2 gives uniform power-decaying partial-sum bounds up to $n^{2-\eta}$. Summation by parts with $1/j$ on the $O(\log k)$ dyadic blocks gives

$$\sum_{n < j \le n^{2-\eta}} \frac{e^{2\mathrm{i} N_k \theta_j}}{j} = o(1).$$

The final strip has absolute weight at most $\eta \log n + O(1)$. Taking $k \to \infty$ first and then $\eta \downarrow 0$ proves that the full harmonic oscillatory sum is $o(\log n)$. Hence the bracket in (30) is $\frac{1}{2} \log n + o(\log n)$. Since $\log n = (\log k)/\alpha$ and $N_k/k \to 1$, we obtain

$$a_k \sim \frac{A D_\rho^2}{2\alpha\tau^\rho} (\log k) k^{-2\rho} = c G_\rho (\log k) k^{-2\rho}.$$

The argument uses only $\rho > 0$ and therefore also covers the newly included boundary $\rho = \omega \le 1$.

4.3 The regime $\omega > \rho$

Let $s = \omega - \rho > 0$ and $n = k^{1/\alpha}$. Define the exact normalized factor

$$Q_k(\lambda) = \frac{p_k(\lambda)}{D_\rho N_k^{-\rho} \lambda^{-\rho/2} q^{(k-1)/2}}.$$

The envelope (18) gives $|Q_k| \le C$ globally, including very small λ. The exact energy becomes

$$a_k = A D_\rho^2 N_k^{-2\rho} \tau^{-\rho} T_k, \quad T_k = \sum_{j \ge 1} j^{-1-\alpha s} q_j^{k-1} Q_k(\lambda_j)^2. \tag{31}$$

On each fixed scaled interval $\varepsilon \le j/n \le R$,

$$Q_k(\lambda_j) = \cos(N_k \theta_j - \rho\pi/2) + O(k^{-1/2}), \qquad q_j^{k-1} = e^{-\tau (j/n)^{-\alpha}} + o(1)$$

uniformly. The nonoscillatory half of $k^s T_k$ therefore tends to

$$\frac{1}{2} \int_\varepsilon^R x^{-1-\alpha s} e^{-\tau x^{-\alpha}}\, dx.$$

Lemma 2 controls the oscillatory half. Its unscaled weight is

$$w_k(y) = \frac{1}{n} (y/n)^{-1-\alpha s} (1 - \tau y^{-\alpha})^{k-1}.$$

On $[\varepsilon n, Rn]$ its supremum and total variation are $O(n^{-1})$: after writing $y = nx$, the remaining function and its first derivative are uniformly bounded on $[\varepsilon, R]$. Summation by parts makes the oscillatory term $o(1)$.

The global bound on Q_k controls the scaled upper tail by $CR^{-\alpha s}$. The lower tail is bounded by

$$\frac{C}{n} \sum_{j/n < \varepsilon} (j/n)^{-1-\alpha s} e^{-(\tau/2)(j/n)^{-\alpha}}.$$

The summand function extends continuously by zero at zero. Thus the lower tail is at most

$$C\varepsilon \sup_{0 < x < \varepsilon} \left\{ x^{-1-\alpha s} e^{-(\tau/2) x^{-\alpha}} \right\} \longrightarrow 0$$

uniformly in n. Both cutoffs can now be removed after the k limit, giving

$$k^s T_k \longrightarrow \frac{1}{2} \int_0^\infty x^{-1-\alpha s} e^{-\tau x^{-\alpha}}\, dx = \frac{\Gamma(s)}{2\alpha \tau^s}.$$

Equation (31) proves the last case of (6). This requires only $s > 0$ and includes $0 < \rho < \omega \le 1$.

4.4 Constant comparison and removable values

The duplication formula gives (16) and converts (5) into Bach's low-regime expression wherever that quotient is defined. The only possible simultaneous poles in the low-regime real-parameter domain come from

$$\frac{\Gamma(z)}{\Gamma(2z)}, \qquad z = 2\rho - \tfrac{1}{2} - \omega > \rho - \tfrac{1}{2} > -\tfrac{1}{2}.$$

They occur at $z = 0$, equivalently $1/4 < \rho < 1/2$ and $\omega = 2\rho - 1/2$. The identity

$$\frac{\Gamma(z)}{\Gamma(2z)} = \frac{\sqrt{\pi}\, 2^{1-2z}}{\Gamma(z + 1/2)}$$

extends continuously with value 2 at zero. Formula (5) uses only positive gamma arguments and supplies exactly this value. There is no exceptional asymptotic regime there. For positive integer ρ the issue does not arise. This completes the proof of Theorem 1.

5 Proof of Theorem 2

Here $\rho = 2$ and $s = \omega - 2 > 0$. Write

$$\lambda_j^0 = \tau j^{-\alpha}, \qquad \theta_j^0 = \arcsin\sqrt{\lambda_j^0}, \qquad C_0 = \frac{\Delta^2}{2\gamma}, \qquad c = \frac{C_0 \tau^{1-\omega}}{\alpha}.$$

We first define one fixed sequence of iteration indices:

$$\begin{aligned} k_1 &= \lceil \max\{16, 4^\alpha\} \rceil\,, \\ k_{m+1} &= \Big\lceil \max\Big\{(m+1)^{4\alpha+4},\ k_m + 1, \\ &\qquad\qquad [4m(m+1)k_m^{1/\alpha} + 10(m+1)]^\alpha\Big\}\Big\rceil. \end{aligned} \tag{32}$$

Put

$$n_m = k_m^{1/\alpha}, \quad A_m = \lceil n_m/m \rceil, \quad B_m = \lfloor m n_m \rfloor, \quad h_m = \frac{\pi}{k_m + 1},$$

and let $\sigma_m = 0$ for even m and $\sigma_m = 1/2$ for odd m. Define the nondecreasing rounding map

$$Q_m(t) = h_m\left(\left\lfloor t/h_m - \sigma_m + \tfrac{1}{2}\right\rfloor + \sigma_m\right), \qquad |Q_m(t) - t| \le h_m/2.$$

With $\mathrm{clamp}(t, l, u) = \max\{l, \min\{t, u\}\}$, set

$$\widetilde{\theta}_j = \mathrm{clamp}\big(Q_m(\theta_j^0), \theta_{B_m+1}^0, \theta_{A_m-1}^0\big) \quad \text{if } A_m \le j \le B_m, \tag{33}$$

and $\widetilde{\theta}_j = \theta_j^0$ off these blocks. Finally put $\widetilde{\lambda}_j = \sin^2\widetilde{\theta}_j$. These definitions are made once; the operator will not change with the running iteration index.

5.1 The fixed grid and its boundaries

The recursion gives $n_1 \geq 4$, $k_m \geq m^{4\alpha+4}$, and

$$n_{m+1} \geq 4m(m+1)n_m + 10(m+1), \qquad A_{m+1} \geq 4B_m + 10.$$

Thus the blocks are separated by unmodified indices and every anchor in (33) is valid. The first block is empty if n_1 is nonintegral, and contains one index otherwise; this harmless convention is included in the definition. For $m \geq 2$ the blocks are nonempty.

The base phases decrease in j, while the rounding and clamping maps are nondecreasing in their arguments. Hence the modified phases are nonincreasing inside each block. The clamp puts its first and last values between the unchanged neighboring phases, so monotonicity also holds across all boundaries. Every phase is positive, and index one remains unchanged. Therefore $\widetilde{\lambda}_j$ is positive and nonincreasing, with $\widetilde{\lambda}_1 = \tau < 1$.

Projection onto the clamping interval cannot increase the error from θ_j^0. On block m,

$$|\widetilde{\theta}_j - \theta_j^0| \leq h_m/2,$$

and the Lipschitz bound for sine gives

$$\left| \frac{\sqrt{\widetilde{\lambda}_j}}{\sqrt{\lambda_j^0}} - 1 \right| \leq \frac{\pi}{2\sqrt{\tau}} \frac{m^{\alpha/2}}{\sqrt{k_m}} \longrightarrow 0. \tag{34}$$

Off the blocks the ratio is one. Every finite collection of blocks contains only finitely many indices, so

$$b_j := \frac{\widetilde{\lambda}_j}{\tau j^{-\alpha}} \longrightarrow 1.$$

In particular $0 < b_* \leq b_j \leq b^* < \infty$ for fixed constants. The operator with diagonal entries $\widetilde{\lambda}_j/\gamma$ is compact and bounded with norm L. Since $\omega > 2$ implies $\beta > 1$, the prescribed initial state is in ordinary ℓ^2.

5.2 The actual criterion and uniform tails

Formula (26) applies at each modified eigenvalue. Keeping that eigenvalue in the objective gives the exact identity

$$a_k = \frac{9C_0}{k^2(k+2)^2} \sum_{j \geq 1} \frac{j^{-\beta}}{\widetilde{\lambda}_j} (1 - \widetilde{\lambda}_j)^{k-1} B_k(\widetilde{\lambda}_j)^2. \tag{35}$$

Put $n = k^{1/\alpha}$ and $x_j = j/n$. Then

$$k^{\omega+2} a_k = \frac{9C_0}{\tau} \left(\frac{k}{k+2} \right)^2 U_k, \tag{36}$$

where

$$U_k = \frac{1}{n} \sum_{j \geq 1} x_j^{-1-\alpha s} b_j^{-1} (1 - \widetilde{\lambda}_j)^{k-1} B_k(\widetilde{\lambda}_j)^2.$$

For $k \geq 2$, (27) bounds the summands by $CF(x_j)/n$, with

$$F(x) = x^{-1-\alpha s} \exp[-(\tau b_*/2) x^{-\alpha}].$$

Uniformly in k, its upper tail beyond R is $O(R^{-\alpha s})$. Its lower tail is at most

$$C\varepsilon \sup_{0<x<\varepsilon} F(x) \longrightarrow 0 \quad \text{as } \varepsilon \downarrow 0.$$

These estimates apply to the entire fixed grid, regardless of which patches meet the tails.

On each compact interval $\varepsilon \leq x_j \leq R$,

$$b_j^{-1} (1 - \widetilde{\lambda}_j)^{k-1} = e^{-\tau x_j^{-\alpha}} + o(1),$$

uniformly. Also $\cot \widetilde{\theta}_j/(k+1) = O(k^{-1/2})$, so

$$B_k(\widetilde{\lambda}_j)^2 = \cos^2((k+1)\widetilde{\theta}_j) + O(k^{-1/2}).$$

Since the cosine square is at most one, compact Riemann sums and the uniform tails give the global bound

$$\limsup_{k \to \infty} U_k \leq I, \qquad I = \int_0^\infty x^{-1-\alpha s} e^{-\tau x^{-\alpha}} \, dx = \frac{\Gamma(s)}{\alpha \tau^s}. \tag{37}$$

This controls all integer iterations, not merely the selected subsequences.

5.3 Two subsequences attain the extrema

Fix $0 < \varepsilon < R$. For large m, every index with $\varepsilon n_m \leq j \leq R n_m$ lies inside block m. The distances from its base phase to both clamp endpoints are at least $C_{\varepsilon,R} k_m^{-1/2}$. Indeed, the upper anchor lies above the base phase at $\varepsilon n_m/2$, and the lower anchor lies below that at $2Rn_m$. The uniform small-angle expansion on this compact scale gives strictly positive limiting gaps after multiplication by $\sqrt{k_m}$. Since the rounding error is $O(k_m^{-1})$, the clamp is inactive there for large m. Thus

$$(k_m + 1)\widetilde{\theta}_j \in \pi(\mathbb{Z} + \sigma_m) \quad \text{throughout every fixed compact scaled interval.} \tag{38}$$

For even m, the sine in (26) is zero and the cosine is ± 1, so $B_{k_m}^2 = 1$ exactly on that interval. For odd m, its cosine is zero and

$$B_{k_m}(\widetilde{\lambda}_j)^2 = \frac{\cot^2 \widetilde{\theta}_j}{(k_m+1)^2} = O(k_m^{-1})$$

uniformly there. Compact Riemann sums, followed by the same uniform tail removal, give

$$U_{k_{2m}} \longrightarrow I, \qquad U_{k_{2m+1}} \longrightarrow 0.$$

Together with nonnegativity, (36) and (37), this proves (8), since $(9C_0/\tau)I = 9c\Gamma(\omega - 2)$.

5.4 The unchanged Stieltjes data

The measure (9) is finite and supported in $[0,\tau] \subset [0,1)$. For $u > 0$, termwise differentiation is justified by its finite mass. For an integer $\ell > \omega$,

$$S^{(\ell-1)}(u) = (-1)^{\ell-1}\Gamma(\ell)C_0 \sum_{j\geq 1} \frac{j^{-\beta}\widetilde{\lambda}_j}{(\widetilde{\lambda}_j + u)^{\ell}}.$$

First use $\lambda_j^0 = \tau j^{-\alpha}$ and the scale $N = (\tau/u)^{1/\alpha}$. A Riemann sum with power-tail bounds gives

$$\begin{aligned}\sum_{j\geq 1} \frac{j^{-\beta}\lambda_j^0}{(\lambda_j^0 + u)^{\ell}} &\sim \tau^{1-\omega}u^{\omega-\ell} \int_0^{\infty} \frac{x^{-1-\alpha\omega}}{(1+x^{-\alpha})^{\ell}}\, dx \\ &= \tau^{1-\omega}u^{\omega-\ell}\frac{\Gamma(\omega)\Gamma(\ell-\omega)}{\alpha\Gamma(\ell)}. \end{aligned} \tag{39}$$

At zero the integrable exponent is $-1+\alpha(\ell-\omega) > -1$, and at infinity it is $-1-\alpha\omega < -1$.

For any $0 < \eta < 1$, one has $1-\eta \leq b_j \leq 1+\eta$ for all sufficiently large j. Uniformly in $u > 0$, the ratio of the actual summand to its baseline version lies between

$$\frac{1-\eta}{(1+\eta)^{\ell}} \quad \text{and} \quad \frac{1+\eta}{(1-\eta)^{\ell}}.$$

Every fixed prefix of either sum is $O(1)$ as $u \downarrow 0$, whereas $u^{\omega-\ell}$ diverges. Apply these bounds to the tail, use (39), and then let $\eta \downarrow 0$. The actual sum has the same equivalent. Multiplying by the derivative factor and using $C_0\tau^{1-\omega}/\alpha = c$ proves (7) for every integer $\ell > \omega$. This completes Theorem 2.

6 Stability under spectral perturbations

We retain the exact initial coordinates but now allow

$$\lambda_j^0 = \tau j^{-\alpha}, \qquad \lambda_j = \lambda_j^0 r_j \in (0,1), \qquad r_j > 0, \quad r_j \longrightarrow 1. \tag{40}$$

Throughout this section a_k uses the actual eigenvalues λ_j, while a_k^0 denotes the criterion for λ_j^0. Thus, with $C_0 = \Delta^2/(2\gamma)$,

$$a_k = C_0 \sum_{j\geq 1} j^{-\beta}\lambda_j p_k(\lambda_j)^2, \qquad c = \frac{C_0\tau^{1-\omega}}{\alpha}.$$

The sequence r_j is bounded above and below, and $\sup_j \lambda_j < 1$. Hence the scalar estimates apply with a fixed upper bound that need not equal the tail coefficient τ.

Theorem 4. *For every fixed $\rho, \alpha, \omega > 0$, the following hold under (40), with $\beta = 1 + \alpha(\omega - 1)$.*

1. *If $0 < \omega < \rho$, then $a_k/a_k^0 \to 1$.*
2. *If $\omega > \rho$ and $r_j - 1 = o(j^{-\alpha/2})$, then $a_k/a_k^0 \to 1$. In particular, $O(j^{-\delta})$ suffices for $\delta > \alpha/2$.*
3. *If $\omega = \rho$ and $r_j - 1 = O(j^{-\alpha/2})$, then*

$$a_k - a_k^0 = O(k^{-2\rho}).$$

In particular, the logarithmic leading constant is unchanged.

No monotonicity, spectral-gap condition, or differentiable interpolation of the perturbed grid is required.

The first statement is a dominated comparison on the Bessel scale. The other two statements distinguish a strict perturbation threshold in the high regime from an inclusive threshold at the logarithmic boundary. We will give matching fixed monotone examples, and an exact sharp interval below the boundary threshold. The proof compares perturbed phases with the exact grid; it does not differentiate an uncontrolled spectral remainder.

6.1 Comparison estimates and proof of Theorem 4

Let $s = \omega - \rho$ and use the exact normalized factor Q_k from Section 4. With $N_k = k + \rho - 1$ and $B_\rho = C_0 D_\rho^2 \tau^{1-\rho}$, direct substitution gives

$$a_k = B_\rho N_k^{-2\rho} \sum_{j \geq 1} j^{-1-\alpha s} r_j^{1-\rho} (1 - \lambda_j)^{k-1} Q_k(\lambda_j)^2. \tag{41}$$

The factor $r_j^{1-\rho}$ comes from the actual objective weights and cannot be omitted.

For $0 < \omega < \rho$, set $m = k^{2/\alpha}$ and $x_j = j/m$. Then

$$k^{2\omega} a_k = \frac{C_0 \tau}{m} \sum_{j \geq 1} x_j^{-1-\alpha\omega} r_j p_k(\tau k^{-2} x_j^{-\alpha} r_j)^2.$$

On any compact interval in $x > 0$, the factors r_j converge uniformly to one, and the Bessel limit from Section 2 is the same as for the exact grid. The scalar envelope bounds the summand, apart from its mesh factor, by

$$C x^{-1-\alpha\omega} \min\{1, x^{\alpha\rho}\}.$$

Its power tails are integrable and uniformly summable exactly when $0 < \omega < \rho$. Dominated Riemann sums prove the first assertion.

For $s > 0$, set $n = k^{1/\alpha}$. On $\varepsilon \le j/n \le R$, the mean value theorem gives

$$|\theta(\lambda_j) - \theta(\lambda_j^0)| \le Cj^{-\alpha/2}|r_j - 1|, \qquad \theta(\lambda) = \arcsin\sqrt{\lambda}. \tag{42}$$

Under the little-o hypothesis, multiplication by N_k makes this phase difference $o(1)$ uniformly there. Also

$$r_j^{1-\rho}(1-\lambda_j)^{k-1} = e^{-\tau(j/n)^{-\alpha}} + o(1), \qquad Q_k(\lambda_j) = \cos(N_k\theta(\lambda_j) - \rho\pi/2) + O(k^{-1/2})$$

uniformly on that interval. The exact-grid cancellation from Lemma 2 therefore gives the same compact-interval limit as before. Since $|Q_k| \le C$, the scaled tails in (41) are bounded by mesh sums of

$$Cx^{-1-\alpha s}e^{-c_1x^{-\alpha}},$$

whose lower and upper tails vanish uniformly. Removing the cutoffs proves the second assertion.

For the boundary it is useful to record a slightly more general reduction. Suppose $\omega = \rho$ and $r_j - 1 = O(j^{-\delta})$ for some $\delta > 0$. The ranges $j \le n$ and $j > n^2$ each contribute $O(k^{-2\rho})$: the first follows from the damped harmonic sum $\sum_{j\le n} j^{-1}e^{-c_1kj^{-\alpha}} = O(1)$, and the second from $\sum_{j>n^2} j^{-1-\alpha\rho} = O(k^{-2\rho})$. On the remaining finite sum, the cosine approximation and its squared remainder cost $O(1)$ inside the prefactor, because

$$k^{-1}\sum_{j\le n^2} j^{\alpha/2-1} = O(1), \qquad k^{-2}\sum_{j\le n^2} j^{\alpha-1} = O(1).$$

Replacing $r_j^{1-\rho}$ by one costs at most $C\sum_{j>n} j^{-1-\delta} = O(1)$, and removing the damping costs at most $Ck\sum_{j>n} j^{-1-\alpha} = O(1)$. Thus

$$a_k = B_\rho N_k^{-2\rho}\left[\sum_{n<j\le n^2} \frac{\cos^2\phi_{kj}}{j} + O(1)\right], \quad \phi_{kj} = N_k\theta(\lambda_j) - \rho\pi/2. \tag{43}$$

Every infinite tail was controlled before this finite-sum replacement. The same identity holds for the reference grid.

In the third assertion, (42) bounds the difference of the two cosine-square sums by

$$Ck\sum_{j>n} j^{-1-\alpha} = O(1).$$

Equation (43) proves the claimed $O(k^{-2\rho})$ difference and finishes Theorem 4.

6.2 Failure at the high-regime threshold

Theorem 5. *Fix $\omega > \rho > 0$ and $\alpha > 0$. There is one positive nonincreasing grid with $\lambda_1 = \tau$ and $\lambda_j = \tau j^{-\alpha}\{1 + O(j^{-\alpha/2})\}$ for which the normalized criterion has two different subsequence limits. More precisely, put*

$$s = \omega - \rho, \quad w_s(x) = x^{-1-\alpha s}e^{-\tau x^{-\alpha}}, \quad \eta_s = \frac{\int_1^2 w_s(x)\,dx}{\int_0^\infty w_s(x)\,dx} \in (0,1).$$

For the integer sequence specified below,

$$k_{2m}^{\omega+\rho} a_{k_{2m}} \longrightarrow cG_\rho \Gamma(s)(1+\eta_s), \tag{44}$$

$$k_{2m+1}^{\omega+\rho} a_{k_{2m+1}} \longrightarrow cG_\rho \Gamma(s)(1-\eta_s). \tag{45}$$

The actual performance measure retains (7) *with the same* c, ω. *If* $\alpha > 2$, *this example also satisfies* $\lambda_j - \lambda_{j+1} \sim \tau\alpha j^{-\alpha-1}$.

Proof. Let

$$k_1 = \lceil \max\{16, 4^\alpha\} \rceil, \qquad n_m = k_m^{1/\alpha}, \quad A_m = \lceil n_m \rceil, \quad B_m = \lfloor 2n_m \rfloor,$$

and define

$$k_{m+1} = \lceil \max\{k_m + 1, [(m+2)(B_m+3)]^\alpha\} \rceil. \tag{46}$$

Then $n_{m+1} \geq (m+2)(B_m+3)$. The blocks and their immediate anchors are disjoint, $B_{m-1}/n_m \to 0$, and $A_{m+1}/n_m \to \infty$.

Here is a phase-rounding rule that we will also use below. Write $\theta_j^0 = \arcsin\sqrt{\tau j^{-\alpha}}$, and set

$$\nu_m = k_m + \rho - 1, \quad h_m = \frac{\pi}{\nu_m}, \quad \sigma_m = \begin{cases} \rho/2, & m \text{ even}, \\ \rho/2 + 1/2, & m \text{ odd}. \end{cases}$$

Define

$$\begin{aligned} R_m(t) &= h_m \left[\left\lfloor t/h_m - \sigma_m + \tfrac{1}{2} \right\rfloor + \sigma_m \right], \\ \theta_j &= \max\{\theta^0_{B_m+1}, \min\{R_m(\theta_j^0), \theta^0_{A_m-1}\}\}, \qquad A_m \leq j \leq B_m. \end{aligned} \tag{47}$$

Off the blocks put $\theta_j = \theta_j^0$, and let $\lambda_j = \sin^2\theta_j$. Rounding and clamping are nondecreasing functions of their argument. Each patched block is therefore nonincreasing and lies between its unchanged neighbors. Since $A_1 \geq 4$, the first eigenvalue is τ; positivity and $\lambda_j \leq \tau$ also follow. Projection cannot increase the distance to θ_j^0, so

$$|\theta_j - \theta_j^0| \leq h_m/2, \qquad \left| \frac{\sin\theta_j}{\sin\theta_j^0} - 1 \right| \leq C\frac{j^{\alpha/2}}{k_m}. \tag{48}$$

On $[A_m, B_m]$, $j \asymp n_m$, proving the single uniform relative bound $O(j^{-\alpha/2})$.

On any fixed compact subinterval of $1 < j/n_m < 2$, the distance from θ_j^0 to each clamp endpoint is of order at least $n_m^{-\alpha/2}$, while $h_m = O(k_m^{-1})$. Hence clamping is eventually inactive. There the squared leading cosine equals one for even m and zero for odd m. The scalar remainder is $O(k_m^{-1/2})$; the damping and actual weight converge to their usual high-scale limits. Edge strips of scaled width ε have contribution $O(\varepsilon)$ after normalization, so they may be removed after taking $m \to \infty$. Past and future blocks lie in the uniformly negligible tails by (46).

Put $I_s = \int_0^\infty w_s = \Gamma(s)/(\alpha\tau^s)$ and $J_s = \int_1^2 w_s$. The reference block contributes $J_s/2$, by Lemma 2; the patched block contributes J_s or zero. Consequently the two limits are

$B_\rho(I_s + J_s)/2$ and $B_\rho(I_s - J_s)/2$. Since $B_\rho I_s/2 = cG_\rho\Gamma(s)$, these are (44)–(45). They are not asserted to be global extrema.

If $\alpha > 2$, each absolute eigenvalue error is $O(j^{-3\alpha/2})$. The error in a consecutive gap is at most the same order, which is $o(j^{-\alpha-1})$. This proves the gap equivalent. Thus first-gap regularity alone does not ensure preservation of the high-regime constant. The A1 assertion follows from the positive comparison at the end of this section. ∎

6.3 The sharp interval below the logarithmic threshold

Theorem 6. *Fix $\omega = \rho > 0$, $\alpha > 0$, and $0 < \delta < \alpha/2$. Put*

$$K = cG_\rho, \qquad b = \frac{1}{\alpha/2 + \delta}, \qquad q_\delta = \alpha b = \frac{2\alpha}{\alpha + 2\delta} \in (1, 2).$$

Every fixed grid (40) *with $r_j - 1 = O(j^{-\delta})$ satisfies*

$$(2 - q_\delta)K \le \liminf_{k\to\infty} \frac{k^{2\rho}a_k}{\log k} \le \limsup_{k\to\infty} \frac{k^{2\rho}a_k}{\log k} \le q_\delta K. \tag{49}$$

No monotonicity or gap assumption is needed for these bounds. There is one fixed positive nonincreasing grid in this class, with $\lambda_1 = \tau$, attaining both outer endpoints as its actual global lower and upper limits. Its actual performance measure retains (7) *with the same c, ρ.*

Proof. First consider an arbitrary admissible grid. Set $n = k^{1/\alpha}$, $J = \lceil k^b \rceil$, and write ϕ^0_{kj} for the reference phase. By (42),

$$\sum_{J\le j\le n^2} \frac{|\cos^2\phi_{kj} - \cos^2\phi^0_{kj}|}{j} \le Ck \sum_{j\ge J} j^{-1-\alpha/2-\delta} = O(1),$$

because $b(\alpha/2 + \delta) = 1$. The exact-grid cancellation lemma, with summation by parts on dyadic blocks, gives

$$\sum_{n<j\le k^b} \frac{e^{2\mathrm{i}N_k\theta^0_j}}{j} = o(1).$$

Indeed $1/\alpha < b < 2/\alpha$, so these blocks stay a fixed power below the upper endpoint used in the boundary proof. The full-window oscillatory sum is $o(\log k)$ by Section 4. Subtraction therefore yields

$$\sum_{J\le j\le n^2} \frac{\cos^2\phi_{kj}}{j} = \frac{1}{2}(2/\alpha - b)\log k + o(\log k).$$

The remaining susceptible strip has the elementary bounds

$$0 \le \sum_{n<j<J} \frac{\cos^2\phi_{kj}}{j} \le (b - 1/\alpha)\log k + O(1).$$

Use (43) and $B_\rho = 2\alpha K$. The lower coefficient is $K(2-\alpha b)$, and the upper coefficient is $K\alpha b$. This proves the universal interval.

For sharpness put $e = b - 1/\alpha > 0$ and define

$$\begin{aligned} k_1 &= \left\lceil \max\{16, 4^\alpha, 8^{1/e}\} \right\rceil, \quad k_{m+1} = k_m^3, \\ n_m &= k_m^{1/\alpha}, \qquad\qquad A_m = \lceil m n_m \rceil, \qquad B_m = \lfloor k_m^b \rfloor. \end{aligned} \tag{50}$$

Apply the same rule (47) to these blocks. Since $k_m = k_1^{3^{m-1}}$, we have $k_m^e \geq 8^{3^{m-1}} \geq 8m$ and $n_m \geq 4$. Every block is nonempty, with positive anchors. Moreover

$$\frac{A_{m+1}}{B_m} \geq (m+1) k_m^{3/\alpha - b} \geq (m+1) n_m.$$

Thus the blocks and their anchors are disjoint. The rounding argument again proves positivity, nonincrease, and $\lambda_1 = \tau$. By (48), on the mth block,

$$j^\delta \left| \frac{\sin\theta_j}{\sin\theta_j^0} - 1 \right| \leq C \frac{j^{\alpha/2+\delta}}{k_m} \leq C.$$

The unweighted relative error tends uniformly to zero along the blocks, since $\alpha b/2 < 1$. Squaring and absorbing finitely many indices proves one global $O(j^{-\delta})$ bound.

At $k = k_m$, the two endpoint scales satisfy

$$\frac{A_m}{n_m} \sim m \to \infty, \qquad \frac{B_m}{n_m^2} \sim k_m^{b-2/\alpha} \to 0.$$

In particular,

$$\sup_{A_m \leq j \leq B_m} k_m \lambda_j \leq C m^{-\alpha} \to 0, \qquad \inf_{A_m \leq j \leq B_m} k_m \sqrt{\lambda_j} \geq c_1 k_m^{1-\alpha b/2} \to \infty. \tag{51}$$

The damping tends uniformly to one, the cosine remainder tends uniformly to zero, and $r_j^{1-\rho} \to 1$ uniformly on the patch.

For $2A_m \leq j \leq B_m/2$, the distance of θ_j^0 to each clamp endpoint is bounded below by a positive multiple of $B_m^{-\alpha/2}$. For the lower endpoint compare the angles at $B_m/2$ and $B_m + 1$; for the upper one compare those at $A_m - 1$ and $2A_m$. Since $k_m^{-1} B_m^{\alpha/2} \to 0$, clamping is inactive there. The squared leading cosine is exactly one for even m and zero for odd m. The two excluded fixed-factor edge strips have harmonic mass $O(1)$, whereas

$$\sum_{A_m \leq j \leq B_m} \frac{1}{j} = e \log k_m - \log m + O(1), \qquad \frac{\log m}{\log k_m} \longrightarrow 0. \tag{52}$$

The reference block has half this leading mass: cancellation applies up to k_m^b, and removing indices from n_m to A_m costs only $O(\log m) = o(\log k_m)$.

All other patched blocks are negligible at this iteration. Indeed,

$$\frac{B_{m-1}}{n_m} \leq k_{m-1}^{b-3/\alpha} \to 0, \qquad \frac{A_{m+1}}{n_m^2} \geq (m+1) k_m^{1/\alpha} \to \infty.$$

They lie in the two $O(k_m^{-2\rho})$ tails of (43), for both grids. Comparing with the exact criterion, (51)–(52) give

$$\frac{k_{2m}^{2\rho} a_{k_{2m}}}{\log k_{2m}} \longrightarrow K + \frac{B_\rho e}{2} = q_\delta K, \qquad \frac{k_{2m+1}^{2\rho} a_{k_{2m+1}}}{\log k_{2m+1}} \longrightarrow K - \frac{B_\rho e}{2} = (2 - q_\delta)K.$$

The universal bounds already proved make these the actual global extrema. The A1 assertion follows from the comparison below. ∎

6.4 Transform data and scope of the perturbation results

Every grid (40), with the exact coordinate powers, has finite initial energy because $\alpha + \beta > 1$. Its diagonal multiplier is compact and bounded, and the scalar envelope gives finite criterion at every iteration. The ordinary ℓ^2 interpretation still requires $\beta > 1$.

The same equivalence of grids preserves the leading Stieltjes data. For an integer $\ell > \omega$, consider the positive sum

$$C_0(\ell - 1)! \sum_j \frac{j^{-\beta}\lambda_j}{(\lambda_j + u)^\ell}.$$

After finitely many indices are discarded, $1 - \varepsilon \le r_j \le 1 + \varepsilon$. Each summand relative to its exact-grid counterpart lies between $(1-\varepsilon)/(1+\varepsilon)^\ell$ and $(1+\varepsilon)/(1-\varepsilon)^\ell$. The discarded sum is $O(1)$, hence negligible against $u^{\omega-\ell}$. The exact-grid equivalent and then $\varepsilon \downarrow 0$ give precisely (7), with the same c, ω. This verifies the actual perturbed masses, without differentiating a tail equivalent.

The examples show failure of a pointwise constant, not a worse algebraic exponent. Their fixed-instance spectral placement has the gluing ancestry already noted in [13]; the constants and perturbation thresholds require the specific phase and damping estimates above. The stability assertions do not claim that every perturbation of critical order changes the limit. They identify sufficient classes and matching fixed examples. The low-regime dominated comparison is supporting Abelian content, while the high/boundary contrast and the sharp interval (49) quantify the phase sensitivity.

7 Proof of Theorem 3

7.1 The actual filter and a temporal-window lower bound

Let $e_k = u^\dagger - u_k$. Direct substitution in (10) gives

$$e_0 = u^\dagger, \qquad e_1 = (I - Q)u^\dagger, \qquad e_{k+1} = (I - Q)\left[e_k + \frac{k-1}{k + 2\rho - 1}(e_k - e_{k-1})\right].$$

Hence $e_k = r_k(Q)u^\dagger$, where $r_0(\lambda) = 1$, $r_1(\lambda) = 1 - \lambda$, and the scalar recurrence has the same coefficients as (13). For $k \ge 1$ its exact residual is

$$r_k(\lambda) = (1 - \lambda)^{(k+1)/2} \frac{C_{k-1}^\rho(\sqrt{1-\lambda})}{C_{k-1}^\rho(1)} = (1 - \lambda)p_k(\lambda). \tag{53}$$

The last equality is only asserted for $k \geq 1$. It gives $r_1 = q$ and $r_2 = q^2$; the coefficient of r_0 in the first recurrence step is zero. In particular, the extra Landweber factor imposed by this initialization is retained.

The proof of Lemma 1 also gives

$$|r_k(\lambda)| \leq C_\rho (1-\lambda)^{(k+1)/2} \min\{1, (k\sqrt{\lambda})^{-\rho}\}, \qquad 0 < \lambda \leq 1. \tag{54}$$

To include $\lambda = 1$, apply the same Hilb formula on $0 \leq \theta \leq \pi/2$, which lies inside its uniform interval $[0, \pi - \epsilon]$ for a fixed $\epsilon > 0$. Multiplication by $(1-\lambda)^{(k+1)/2}$ gives $r_k(1) = 0$. The value at zero is $r_k(0) = 1$. The normalized Bessel endpoint remains valid for every $\rho > 0$, including $\rho < 1/2$. The finitely many small degrees are absorbed into C_ρ.

Lemma 3. *For every fixed $\rho > 0$, there exist $c_\rho > 0$ and an integer $N_0(\rho)$ such that*

$$\frac{1}{N+1} \sum_{k=N}^{2N} |r_k(\lambda)|^2 \geq c_\rho N^{-\rho} \tag{55}$$

for every integer $N \geq N_0(\rho)$ and every $\lambda \in [1/N, 2/N]$.

Proof. For $N \leq k \leq 2N$ and $\lambda \in [1/N, 2/N]$, put $\theta = \arcsin\sqrt{\lambda}$. The uniform shrinking-angle estimate in Lemma 1, together with (53), gives

$$r_k(\lambda) = A_{k,\lambda} \left[\cos((k+\rho-1)\theta - \rho\pi/2) + \eta_k(\lambda)\right], \qquad |\eta_k(\lambda)| \leq C_\rho N^{-1/2}, \tag{56}$$

where the positive amplitude is

$$A_{k,\lambda} = D_\rho (1-\lambda)^{(k+1)/2} (k+\rho-1)^{-\rho} \lambda^{-\rho/2}.$$

Uniformly on the whole window,

$$c_1 N^{-\rho} \leq A_{k,\lambda}^2 \leq C_1 N^{-\rho}. \tag{57}$$

For the lower bound, $(1-\lambda)^{k+1}$ must not be dropped. It is bounded below by a fixed positive constant on this range: for large N, $\log(1-\lambda) \geq -\lambda/(1-\lambda)$ and $(k+1)\lambda/(1-\lambda)$ is bounded. The other factors in (57) follow from $k + \rho - 1 \asymp N$ and $\lambda \asymp N^{-1}$.

The doubled phase is linear in k. A finite geometric series yields

$$\left| \sum_{k=N}^{2N} e^{2\mathrm{i}k\theta} \right| \leq \frac{1}{\sin\theta} \leq \sqrt{N}.$$

Consequently, uniformly in λ,

$$\frac{1}{N+1} \sum_{k=N}^{2N} \cos^2((k+\rho-1)\theta - \rho\pi/2) = \frac{1}{2} + O(N^{-1/2}). \tag{58}$$

Use $(a+b)^2 \geq a^2/2 - b^2$ in (56), and then use both sides of (57). The averaged positive term is a fixed positive multiple of $N^{-\rho}$ by (58); the subtracted term is $O(N^{-\rho-1})$. Enlarging N_0 proves (55). ∎

This lemma does not require a lower bound at a single iteration or exclude a zero of an individual residual. It uses a finite average of the exact spectral identities under an all-iteration rate hypothesis; it does not change or average the computed algorithm.

7.2 Necessity of the weak spectral tail

Assume $\|r_k(Q)u^\dagger\| \le Ck^{-\xi}$ for all sufficiently large k. For large integers N, spectral calculus and Lemma 3 give

$$\begin{aligned} C^2N^{-2\xi} &\ge \frac{1}{N+1}\sum_{k=N}^{2N}\|r_k(Q)u^\dagger\|^2 \\ &= \int_{[0,1]}\frac{1}{N+1}\sum_{k=N}^{2N}|r_k(\lambda)|^2\,d\mu(\lambda) \\ &\ge c_\rho N^{-\rho}\mu([1/N,2/N]). \end{aligned}$$

Only a finite sum is interchanged with the positive spectral integral. With $s = 2\xi - \rho > 0$, it follows that

$$\mu([1/N,2/N]) \le C'N^{-s}.$$

Apply this at dyadic $N = 2^m$. These annuli cover the small positive spectrum, and their bounds form a summable geometric series. Thus

$$\mu((0,t]) = O(t^s).$$

Possible endpoint overlap only enlarges this upper bound. The atom at zero is absent because $u^\dagger \perp \ker T$. This proves necessity in (11).

7.3 Sufficiency for arbitrary spectral measures

Suppose $F(t) = \mu((0,t]) = O(t^s)$ with $s = 2\xi - \rho > \rho$. Increase the constant so that $F(t) \le Ct^s$ for every $t > 0$, extending F constantly beyond one. From (54),

$$|r_k(\lambda)|^2 \le C_\rho k^{-2\rho}\lambda^{-\rho}e^{-k\lambda}, \qquad 0 < \lambda \le 1.$$

Let $g_k(t) = t^{-\rho}e^{-kt}$. Since $g_k(\lambda) = \int_\lambda^\infty [-g_k'(t)]\,dt$ and the integrands are nonnegative, Tonelli's theorem gives

$$\begin{aligned} \int_{(0,1]} g_k(\lambda)\,d\mu(\lambda) &= \int_0^\infty F(t)\left(\rho t^{-\rho-1} + kt^{-\rho}\right)e^{-kt}\,dt \\ &\le C\{\rho\Gamma(s-\rho) + \Gamma(s-\rho+1)\}k^{-(s-\rho)}. \end{aligned}$$

Both gamma arguments are positive because $s > \rho$. Hence

$$\|r_k(Q)u^\dagger\|^2 \le Ck^{-2\rho-(s-\rho)} = Ck^{-(s+\rho)} = Ck^{-2\xi},$$

which proves sufficiency. No monotonicity of the signed residual is used. A spectral atom at one is harmless because $r_k(1) = 0$ for $k \ge 1$.

The strict inequality $\xi > \rho$ is used here as $s > \rho$. The displayed Gamma integral diverges at equality; Section 8 treats that endpoint by a different window argument. No conclusion about stopping indices selected in the presence of noise follows here.

7.4 Classical range consequence and endpoint failure

For $0 < \kappa < s/2$, the tail bound gives

$$\int_{(0,1]} \lambda^{-2\kappa}\, d\mu(\lambda) < \infty$$

by a dyadic annulus sum: its terms are bounded by a constant times $2^{-m(s-2\kappa)}$. Spectral calculus therefore defines $w = Q^{-\kappa}u^\dagger \in X$ and $u^\dagger = Q^\kappa w$. Since $s/2 = \xi - \rho/2$, this proves (12). This tail-to-range passage and its interpolation-space interpretation are classical [1, 2].

For the endpoint distinction, take $X = Y = \ell^2$,

$$Qe_j = 2^{-j}e_j, \qquad T = Q^{1/2}, \qquad u_j^\dagger = 2^{-js/2}, \qquad s = 2\xi - \rho > \rho.$$

The positive geometric sum gives $F(t) \le Ct^s$, so the sufficient direction already proved yields the all-iteration rate $O(k^{-\xi})$ for this fixed example. However, $Q^{-s/2}u^\dagger$ has every coordinate equal to one and does not belong to ℓ^2. Thus $u^\dagger \notin \operatorname{Ran} Q^{s/2}$. This establishes the endpoint assertion and completes Theorem 3.

8 The exact semisaturation endpoint

We keep the Hilbert-space operator, minimum-norm solution, exact data, and zero-start iteration (10). In particular, $\mu(B) = \|E_Q(B)u^\dagger\|^2$ is finite and $\mu(\{0\}) = 0$. For fixed $\rho > 0$, define the locally finite measure on $(0, 1]$

$$d\nu(\lambda) = \lambda^{-\rho}\, d\mu(\lambda).$$

Its total mass is allowed to be infinite.

Theorem 7. *For every fixed $\rho > 0$, the following are equivalent:*

1. *The exact iteration (10) satisfies $\|u_k - u^\dagger\| = O(k^{-\rho})$ at every sufficiently large integer k.*

2. *For some fixed $t_0 \in (0, 1)$,*

$$\sup_{0<t<t_0} \int_{(t^2,t]} \lambda^{-\rho}\, d\mu(\lambda) < \infty. \tag{59}$$

The condition is unchanged if it is required for every $t_0 \in (0, 1)$, or if $(t^2, t]$ is replaced by $(At^2, Bt]$ for any fixed $A, B > 0$ and all sufficiently small t. It implies $\mu((0, t]) = O(t^\rho)$. Strong membership $u^\dagger \in \operatorname{Ran} Q^{\rho/2}$ implies (59), and both this inclusion and the inclusion into the weak-tail class are strict.

The operator need not be compact or injective, and its spectral measure need not have a density or prescribed grid. This is an exact-data, zero-start, all-iteration characterization. It does not replace the endpoint by a weak-tail condition or continue the divergent integral in Section 7 to equality. The condition measures weighted mass on moving logarithmic spectral intervals; no new general interpolation-space theory is claimed.

8.1 A lower bound across the full spectral window

Lemma 4. *For every fixed $\rho > 0$, there are $c_\rho > 0$ and an integer $N_0(\rho)$ such that*

$$\frac{1}{N+1}\sum_{k=N}^{2N} |r_k(\lambda)|^2 \geq c_\rho N^{-2\rho}\lambda^{-\rho}, \qquad N^{-2} \leq \lambda \leq N^{-1}, \quad N \geq N_0(\rho). \tag{60}$$

Proof. Set $z = N\sqrt{\lambda} \in [1, \sqrt{N}]$ and $\theta = \arcsin\sqrt{\lambda}$. First consider $R \leq z \leq \sqrt{N}$, where the fixed R will be chosen large. Lemma 1 and (53) give the cosine expansion of (56), with its stated amplitude and the new bound $|\eta_k| \leq C_\rho/z$. Uniformly over $N \leq k \leq 2N$,

$$a_\rho z^{-2\rho} \leq A_{k,\lambda}^2 \leq b_\rho z^{-2\rho}.$$

The lower bound retains $(1-\lambda)^{k+1} \geq (1-1/N)^{2N+1}$, which stays bounded away from zero. The exact geometric sum gives

$$\left|\frac{1}{N+1}\sum_{k=N}^{2N} e^{2\mathrm{i}(k+\rho-1)\theta}\right| \leq \frac{1}{(N+1)\sin\theta} \leq \frac{1}{z}.$$

For $z \geq 4$, the mean of the squared leading cosine is at least $3/8$. Using $(a+b)^2 \geq a^2/2 - b^2$ and the two amplitude bounds gives

$$\frac{1}{N+1}\sum_{k=N}^{2N} |r_k(\lambda)|^2 \geq z^{-2\rho}\left(\frac{3a_\rho}{16} - \frac{b_\rho C_\rho^2}{z^2}\right).$$

Choose R so that the bracket is positive.

It remains to include the compact Bessel transition $1 \leq z \leq R$, where a large-phase approximation is insufficient. Since $\lambda = z^2/N^2 = ((k/N)z)^2/k^2$, the compact limit in Lemma 1 and $r_k = (1-\lambda)p_k$ imply

$$r_k(z^2/N^2) = \Phi_\rho((k/N)z) + o(1)$$

uniformly for $N \leq k \leq 2N$ and $1 \leq z \leq R$. Uniform Riemann sums therefore give

$$\frac{1}{N+1}\sum_{k=N}^{2N} |r_k(z^2/N^2)|^2 \longrightarrow \int_1^2 \Phi_\rho(vz)^2\, dv \tag{61}$$

uniformly on $[1, R]$. The integral is continuous and strictly positive. The normalized Bessel power series is analytic with value one at zero. If the integral vanished at some $z > 0$, that profile would vanish on $[z, 2z]$, contradicting $\Phi_\rho(0) = 1$. Its minimum on $[1, R]$ is consequently positive. For sufficiently large N, the left side of (61) is bounded below by a positive constant, hence by that constant times $z^{-2\rho}$. Combining the two ranges proves (60). This argument includes every individual Bessel zero and every $\rho > 0$. ∎

8.2 Necessity and the weak-tail implication

Assume the all-iteration rate in Theorem 7. Integrating (60) and interchanging only a finite sum with the positive spectral integral yield

$$CN^{-2\rho} \geq \frac{1}{N+1} \sum_{k=N}^{2N} \|r_k(Q)u^\dagger\|^2 \geq c_\rho N^{-2\rho} \nu([N^{-2}, N^{-1}]).$$

Thus these closed-window masses are uniformly bounded. For $2^{-m-1} \leq t \leq 2^{-m}$, the interval $(t^2, t]$ is contained in

$$[2^{-2m}, 2^{-m}] \ \cup \ [2^{-2m-2}, 2^{-m-1}].$$

The two intervals overlap for $m \geq 1$ and correspond to integer $N = 2^m, 2^{m+1}$. They cover all atoms at their endpoints as well. This proves (59).

Conversely suppose that condition holds, with bound M. The disjoint intervals $(t^{2^{h+1}}, t^{2^h}]$, $h \geq 0$, partition $(0, t]$. On each interval $\lambda^\rho \leq t^{\rho 2^h}$, so

$$\mu((0,t]) \leq M \sum_{h \geq 0} t^{\rho 2^h} \leq \frac{Mt^\rho}{1-t^\rho} = O(t^\rho). \tag{62}$$

This proves the weak-tail assertion and supplies the low-frequency control needed below; no additional tail assumption is imposed.

8.3 Sufficiency and the two omitted spectral bands

The actual residual bound is

$$|r_k(\lambda)|^2 \leq C_\rho e^{-k\lambda} \min\{1, k^{-2\rho}\lambda^{-\rho}\}, \qquad 0 < \lambda \leq 1.$$

Split its spectral integral at k^{-2} and k^{-1}. The part below k^{-2} is $O(k^{-2\rho})$ by (62). The middle part is at most $C_\rho k^{-2\rho} \nu((k^{-2}, k^{-1}]) = O(k^{-2\rho})$.

For the upper part, it suffices to bound $\int_{(k^{-1},1]} e^{-k\lambda}\, d\nu(\lambda)$ uniformly in k. Choose $t_1 < \min\{t_0, 1/2\}$. A dyadic shell $(2^h/k, 2^{h+1}/k]$, with upper endpoint $v \leq t_1$, lies in $(v^2, v]$ because $v/2 \geq v^2$. Its ν-mass is at most M and its damping at most e^{-2^h}. Their sum is bounded by $M \sum_{h \geq 0} e^{-2^h} < \infty$. The remaining portion lies in $[t_1/2, 1]$, whose ν-mass is finite because μ is finite. The upper spectral band is therefore also $O(k^{-2\rho})$. All three pieces prove the claimed norm rate.

The minimum-norm hypothesis is essential here: a nonzero atom at zero would have residual one at every iteration, while the annulus condition would not detect it. A finite atom at one is harmless because $r_k(1) = 0$ for $k \geq 1$.

8.4 Endpoint conventions and strict inclusions

Changing $t_0 \in (0,1)$ affects only a compact interval away from zero, where ν is finite. To check fixed multiplicative endpoints, push ν forward under $x = -\log\lambda$ to a measure ζ and put $s = \log(1/t)$. The original condition is bounded mass on $[s, 2s)$. The interval $(At^2, Bt]$ becomes $[s - \log B, 2s - \log A)$. For large s it lies strictly inside $(s/2, 4s)$, covered by the three half-open original windows $[s/2, s)$, $[s, 2s)$, $[2s, 4s)$. Hence the original condition implies the modified one, including all atoms.

For the reverse implication put $C = A/B^2$. The modified window at t/B is $(Ct^2, t]$. If $C \le 1$ it covers $(t^2, t]$. If $C > 1$, the additional modified window at Ct^2/B is $(C^3t^4, Ct^2]$; for sufficiently small t, these two windows cover $(t^2, t]$ without losing their common endpoint. This proves the asserted equivalence of conventions.

Finally, spectral calculus identifies $u^\dagger \in \operatorname{Ran} Q^{\rho/2}$ with $\nu((0,1]) < \infty$, since the nullspace component is absent. This implies the annulus condition. To see that both inclusions are strict, take $X = Y = \ell^2$, $Qe_j = 2^{-j}e_j$, and $T = Q^{1/2}$. For $|u_j^\dagger|^2 = 2^{-\rho j}/j$, the weighted atoms are $1/j$. Writing $t = 2^{-L}$ gives

$$\nu((t^2, t]) = \sum_{L \le j < 2L} \frac{1}{j} \le 2 \qquad (L \ge 1),$$

whereas the total harmonic sum diverges. Thus the endpoint rate holds without the strong source membership.

For $|u_j^\dagger|^2 = 2^{-\rho j}$, a geometric sum gives $\mu((0,t]) \le t^\rho/(1-2^{-\rho})$, while $\nu((2^{-2m}, 2^{-m}]) = m$. The annulus condition fails. More directly, (60) at $N = 2^m$ gives an average squared error at least $c_\rho N^{-2\rho}(m+1)$, so $k^\rho\|u_k - u^\dagger\|$ is unbounded along a subsequence. Both vectors are in ℓ^2 and remain fixed throughout the iteration. This completes Theorem 7.

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